# Lab – 04

## ➔ Implement SQL Functions: Conversion Functions, and Date Functions.

```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);

INSERT INTO STUDENT VALUES
(101, 'Aarav Sharma', '10th', 485, 25000.00, '2008-05-14', '2022-04-10'),
(102, 'Ananya Verma', '10th', 462, 25000.00,'2008-09-22', '2022-04-12'),
(103, 'Rohan Mehta', '9th', 390, 18500.50, '2009-01-30', '2023-04-05'),
(104, 'Priya Singh', '11th', 495, 30000.00, '2007-11-08', '2021-06-15'),
(105, 'Kabir Nair', '9th', 415, 20000.00, '2009-07-19', '2023-04-18');
SELECT  * FROM  Student;
```

SQL-functions.sql

```
CREATE TABLE STUDENT (
    Student_ID INT,
    Student_Name VARCHAR(50),
    Class VARCHAR(10),
    Marks INT,
    Fee_Paid DECIMAL(10,2),
    Date_of_Birth DATE,
    Admission_Date DATE
);

INSERT INTO STUDENT VALUES
(101, 'Aarav Sharma', '10th', 485, 25000.00, '2008-05-14', '2022-04-10'),
(102, 'Ananya Verma', '10th', 462, 25000.00, '2008-09-22', '2022-04-12'),
(103, 'Rohan Mehta', '9th', 390, 18500.50, '2009-01-30', '2023-04-05'),
(104, 'Priya Singh', '11th', 495, 30000.00, '2007-11-08', '2021-06-15'),
(105, 'Kabir Nair', '9th', 415, 20000.00, '2009-07-19', '2023-04-18');
SELECT * FROM Student;
```

Output:

| Student_ID | Student_Name | Class | Marks | Fee_Paid | Date_of_Birth | Admission_Date |
|---|---|---|---|---|---|---|
| 101 | Aarav Sharma | 10th | 485 | 25000.00 | 2008-05-14 | 2022-04-10 |
| 102 | Ananya Verma | 10th | 462 | 25000.00 | 2008-09-22 | 2022-04-12 |
| 103 | Rohan Mehta | 9th | 390 | 18500.50 | 2009-01-30 | 2023-04-05 |
| 104 | Priya Singh | 11th | 495 | 30000.00 | 2007-11-08 | 2021-06-15 |
| 105 | Kabir Nair | 9th | 415 | 20000.00 | 2009-07-19 | 2023-04-18 |

```
/* Common String Functions */
SELECT Student_Name,
```

```
    UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
       LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

SELECT Student_Name,
       LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
       SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

SELECT
    Student_Name AS Original_Name,
    TRIM(Student_Name) AS Cleaned_Name
FROM STUDENT;
```

```sql

/* Common String Functions */
SELECT Student_Name,
    UPPER(Student_Name) AS Uppercase_Name
FROM STUDENT;

SELECT Student_Name,
        LOWER(Student_Name) AS Lowercase_Name
FROM STUDENT;

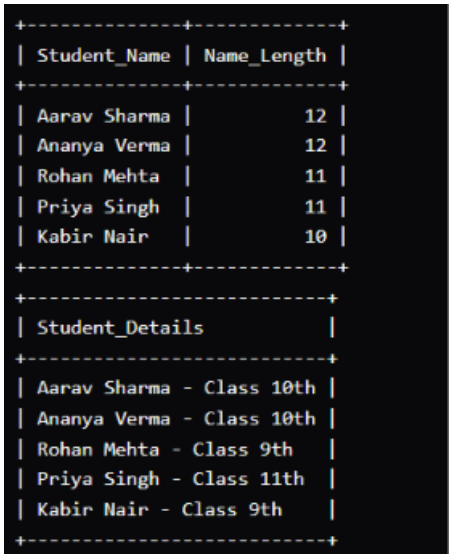
SELECT Student_Name,
        LENGTH(Student_Name) AS Name_Length
FROM STUDENT;

SELECT CONCAT(Student_Name, ' - Class ', Class) AS Student_Details
FROM STUDENT;

SELECT Student_Name,
        SUBSTRING(Student_Name, 1, 5) AS First_Five_Characters
FROM STUDENT;

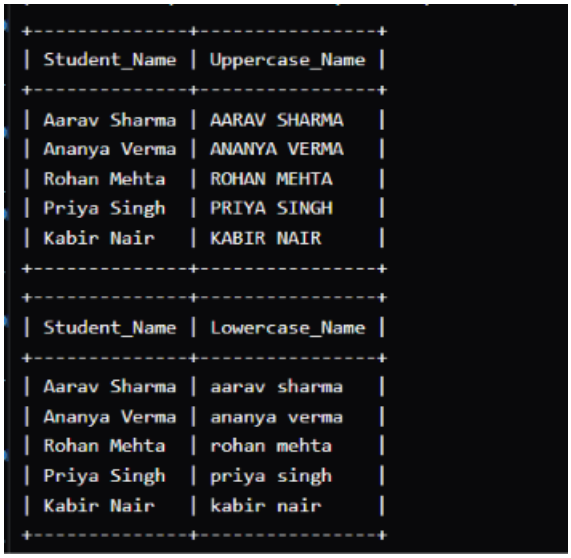
SELECT
    Student_Name AS Original_Name,
    TRIM(Student_Name) AS Cleaned_Name
FROM STUDENT;

```



| Student_Name | Name_Length |
|---|---|
| Aarav Sharma | 12 |
| Ananya Verma | 12 |
| Rohan Mehta | 11 |
| Priya Singh | 11 |
| Kabir Nair | 10 |

| Student_Details |
|---|
| Aarav Sharma - Class 10th |
| Ananya Verma - Class 10th |
| Rohan Mehta - Class 9th |
| Priya Singh - Class 11th |
| Kabir Nair - Class 9th |



| Student_Name | Uppercase_Name |
|---|---|
| Aarav Sharma | AARAV SHARMA |
| Ananya Verma | ANANYA VERMA |
| Rohan Mehta | ROHAN MEHTA |
| Priya Singh | PRIYA SINGH |
| Kabir Nair | KABIR NAIR |

| Student_Name | Lowercase_Name |
|---|---|
| Aarav Sharma | aarav sharma |
| Ananya Verma | ananya verma |
| Rohan Mehta | rohan mehta |
| Priya Singh | priya singh |
| Kabir Nair | kabir nair |



| Student_Name | First_Five_Characters |
|---|---|
| Aarav Sharma | Aarav |
| Ananya Verma | Anany |
| Rohan Mehta | Rohan |
| Priya Singh | Priya |
| Kabir Nair | Kabir |

| Original_Name | Cleaned_Name |
|---|---|
| Aarav Sharma | Aarav Sharma |
| Ananya Verma | Ananya Verma |
| Rohan Mehta | Rohan Mehta |
| Priya Singh | Priya Singh |
| Kabir Nair | Kabir Nair |

```
/* Numeric Functions */
SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;
SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
```

FROM STUDENT;

```sql
/* Numeric Functions */
SELECT Student_Name,
       Marks,
       ROUND(Marks * 1.05) AS Bonus_Marks
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       CEIL(Marks / 10.0) AS Rounded_Up
FROM STUDENT;

SELECT Student_Name,
       Marks / 10.0 AS Marks_Value,
       FLOOR(Marks / 10.0) AS Rounded_Down
FROM STUDENT;

SELECT Student_Name,
       Marks,
       MOD(Marks, 10) AS Remainder
FROM STUDENT;

SELECT Student_Name,
       Marks,
       ABS(Marks - 80) AS Difference_From_80
FROM STUDENT;
```

| Student_Name | Marks | Bonus_Marks |
|---|---|---|
| Aarav Sharma | 485 | 509 |
| Ananya Verma | 462 | 485 |
| Rohan Mehta | 390 | 410 |
| Priya Singh | 495 | 520 |
| Kabir Nair | 415 | 436 |

| Student_Name | Marks_Value | Rounded_Up |
|---|---|---|
| Aarav Sharma | 48.5000 | 49 |
| Ananya Verma | 46.2000 | 47 |
| Rohan Mehta | 39.0000 | 39 |
| Priya Singh | 49.5000 | 50 |
| Kabir Nair | 41.5000 | 42 |

| Student_Name | Marks_Value | Rounded_Down |
|---|---|---|
| Aarav Sharma | 48.5000 | 48 |
| Ananya Verma | 46.2000 | 46 |
| Rohan Mehta | 39.0000 | 39 |
| Priya Singh | 49.5000 | 49 |
| Kabir Nair | 41.5000 | 41 |

| Student_Name | Marks | Remainder |
|---|---|---|
| Aarav Sharma | 485 | 5 |
| Ananya Verma | 462 | 2 |
| Rohan Mehta | 390 | 0 |
| Priya Singh | 495 | 5 |
| Kabir Nair | 415 | 5 |

| Student_Name | Marks | Difference_From_80 |
|---|---|---|
| Aarav Sharma | 485 | 405 |
| Ananya Verma | 462 | 382 |
| Rohan Mehta | 390 | 310 |
| Priya Singh | 495 | 415 |
| Kabir Nair | 415 | 335 |

/* Date Functions */
SELECT Student_Name,
Date_of_Birth,

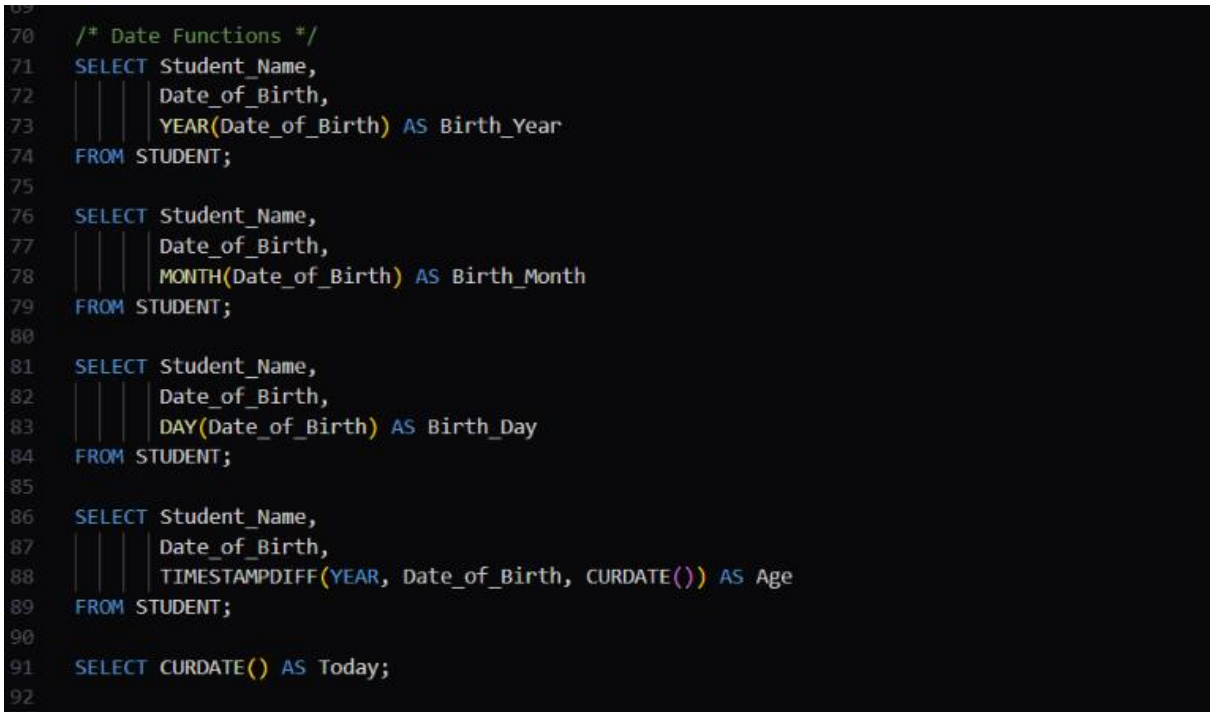
```
    YEAR(Date_of_Birth) AS Birth_Year
FROM STUDENT;

SELECT Student_Name,
    Date_of_Birth,
    MONTH(Date_of_Birth) AS Birth_Month
FROM STUDENT;

SELECT Student_Name,
    Date_of_Birth,
    DAY(Date_of_Birth) AS Birth_Day
FROM STUDENT;

SELECT Student_Name,
    Date_of_Birth,
    TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE()) AS Age
FROM STUDENT;

SELECT CURDATE() AS Today;
```



| Student_Name | Date_of_Birth | Birth_Day |
|---|---|---|
| Aarav Sharma | 2008-05-14 | 14 |
| Ananya Verma | 2008-09-22 | 22 |
| Rohan Mehta | 2009-01-30 | 30 |
| Priya Singh | 2007-11-08 | 8 |
| Kabir Nair | 2009-07-19 | 19 |

| Student_Name | Date_of_Birth | Age |
|---|---|---|
| Aarav Sharma | 2008-05-14 | 18 |
| Ananya Verma | 2008-09-22 | 17 |
| Rohan Mehta | 2009-01-30 | 17 |
| Priya Singh | 2007-11-08 | 18 |
| Kabir Nair | 2009-07-19 | 17 |

| Student_Name | Date_of_Birth | Birth_Year |
|---|---|---|
| Aarav Sharma | 2008-05-14 | 2008 |
| Ananya Verma | 2008-09-22 | 2008 |
| Rohan Mehta | 2009-01-30 | 2009 |
| Priya Singh | 2007-11-08 | 2007 |
| Kabir Nair | 2009-07-19 | 2009 |

| Student_Name | Date_of_Birth | Birth_Month |
|---|---|---|
| Aarav Sharma | 2008-05-14 | 5 |
| Ananya Verma | 2008-09-22 | 9 |
| Rohan Mehta | 2009-01-30 | 1 |
| Priya Singh | 2007-11-08 | 11 |
| Kabir Nair | 2009-07-19 | 7 |

```
/* AGGREGATE FUNCTIONS */
SELECT COUNT(*) AS Total_Students
FROM STUDENT;

SELECT SUM(Marks) AS Total_Marks
FROM STUDENT;

SELECT AVG(Marks) AS Average_Marks
FROM STUDENT;

SELECT MAX(Marks) AS Highest_Marks
FROM STUDENT;

SELECT MIN(Marks) AS Lowest_Marks
FROM STUDENT;
```

```sql

    /* AGGREGATE FUNCTIONS */
    SELECT COUNT(*) AS Total_Students
    FROM STUDENT;

    SELECT SUM(Marks) AS Total_Marks
    FROM STUDENT;

    SELECT AVG(Marks) AS Average_Marks
    FROM STUDENT;

    SELECT MAX(Marks) AS Highest_Marks
    FROM STUDENT;

    SELECT MIN(Marks) AS Lowest_Marks
    FROM STUDENT;

```

Output:

```
+----------------+
| Total_Students |
+----------------+
|              5 |
+----------------+
+-------------+
| Total_Marks |
+-------------+
|        2247 |
+-------------+
+---------------+
| Average_Marks |
+---------------+
|      449.4000 |
+---------------+
+---------------+
| Highest_Marks |
+---------------+
|           495 |
+---------------+
+--------------+
| Lowest_Marks |
+--------------+
|          390 |
+--------------+
```